B. Tech
Engg. Physics

ONE SHOT Revision

UNIT-4 : Laser

Notes + PYQs + DPP

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GATEWAY CLASSES

AKTU

B. Tech I-Year

Gateway Classes

BAS101 / BAS201: ENGINEERING PHYSICS

Unit-4

Introduction to : Fiber Optics & Laser

Syallbus

Fibre Optics: Principle and construction of optical fiber, Acceptance angle, Numerical aperture, Acceptance cone, Step index and graded index fibers, Fiber optic communication principle, Attenuation, Dispersion, Application of fiber.

Laser: Absorption of radiation, Spontaneous and stimulated emission of radiation, Population inversion, Einstein's Coefficients, Principles of laser action, Solid state Laser (Ruby laser) and Gas Laser (He-Ne laser), Laser applications.

LASER

- The word "LASER" is an acronym for LIGHT AMPLIFICATION BY STIMULATED EMISSION OF RADIATION
- Laser depends on the phenomenon of "Stimulated Emission".
- In 1917, Einstein gave the concept of "Stimulated Emission".
- Laser are invented and developed between 1959 and 1962.
- In 1960, Dr. T.H. Maiman invented first working LASER known as "RUBY LASER".

Properties of LASER

(i) Highly Monochromatic

- A LASER emits light beam of single frequency and single wavelength (single color).

(ii) Highly Intense

- All the power or energy is concentrated with in a small area

(iii) Highly Coherent

- The constituent waves are exactly in the same phase.

(iv) Unidirectional

- A laser emits light beam in one particular direction.

(v) Collimated

- All rays are parallel to each other and do not diverge significantly even over long distances.

Thus, Laser is device which emits a highly monochromatic, highly intense, highly coherent, unidirectional and collimated beam of light

Explan the following mechanism

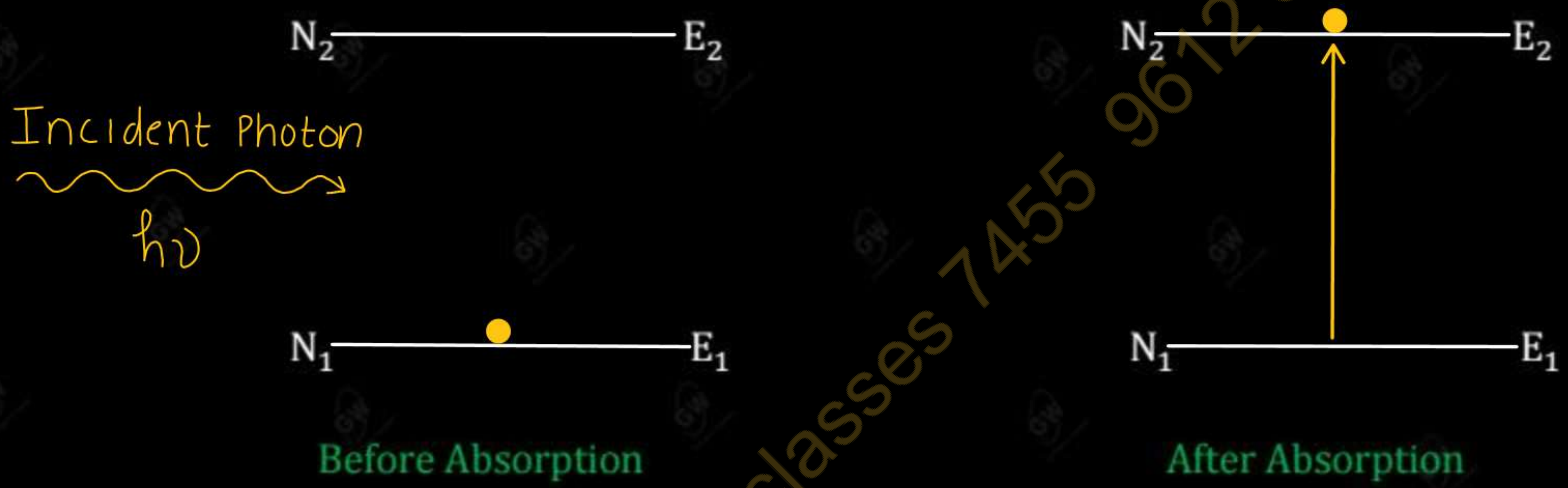
(i) Stimulated Absorption

(ii) Spontaneous Emission

(iii) Stimulated Emission

Stimulated Absorption

- Let us consider two energy level 1 and 2 of an atom with energies E_1 and E_2 as shown below



- An atom residing in the lower energy state E_1 may absorb the incident photon and jump to the higher energy state or excited state E_2, provided

$$E_2 - E_1 = h\upsilon$$

- This process is called stimulated absorption or induced absorption or absorption of radiation

- The rate of absorption is directly proportional to the (i) number of atoms in lower energy state 1 (N_1) and (ii) energy density $E(\upsilon)$

$$P_{12} \propto N_1$$

$$P_{12} \propto E(\upsilon)$$

$$P_{12} \propto N_1 E(\upsilon)$$

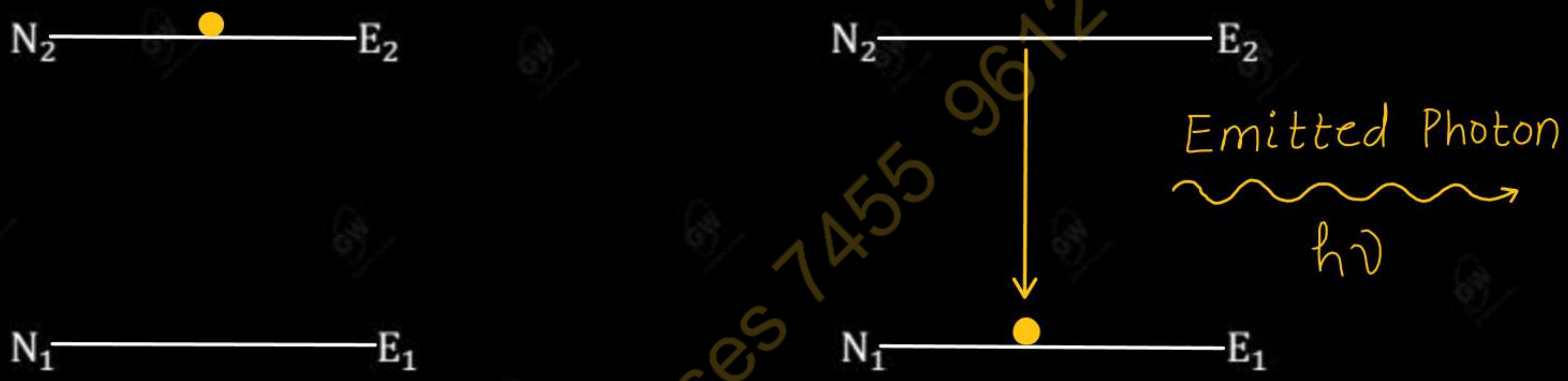
$$\boxed{P_{12} = N_1 B_{12} E(\upsilon)}$$

Where

B_{12} = Einstein's coefficient of absorption

Spontaneous Emission

- Let us suppose that the atom is in higher energy state or excited state E_2



Before spontaneous Emission

After spontaneous Emission

- The excited atom in the state E_2 returns to lower state E_1 to attain stability
- The mean life of atom in excited state is 10^{-8} seconds.
- During this transition atom will release a photon of energy hυ such that

$$E_2 - E_1 = h\upsilon$$

- This type of process in which photon emission occurs without any interaction with external radiation is called spontaneous emission.
- The rate of spontaneous emission is directly proportional to the number of atoms in excited state (N_2)

$$P'_{21} \propto N_2$$

$$\boxed{P'_{21} = N_2 A_{21}}$$

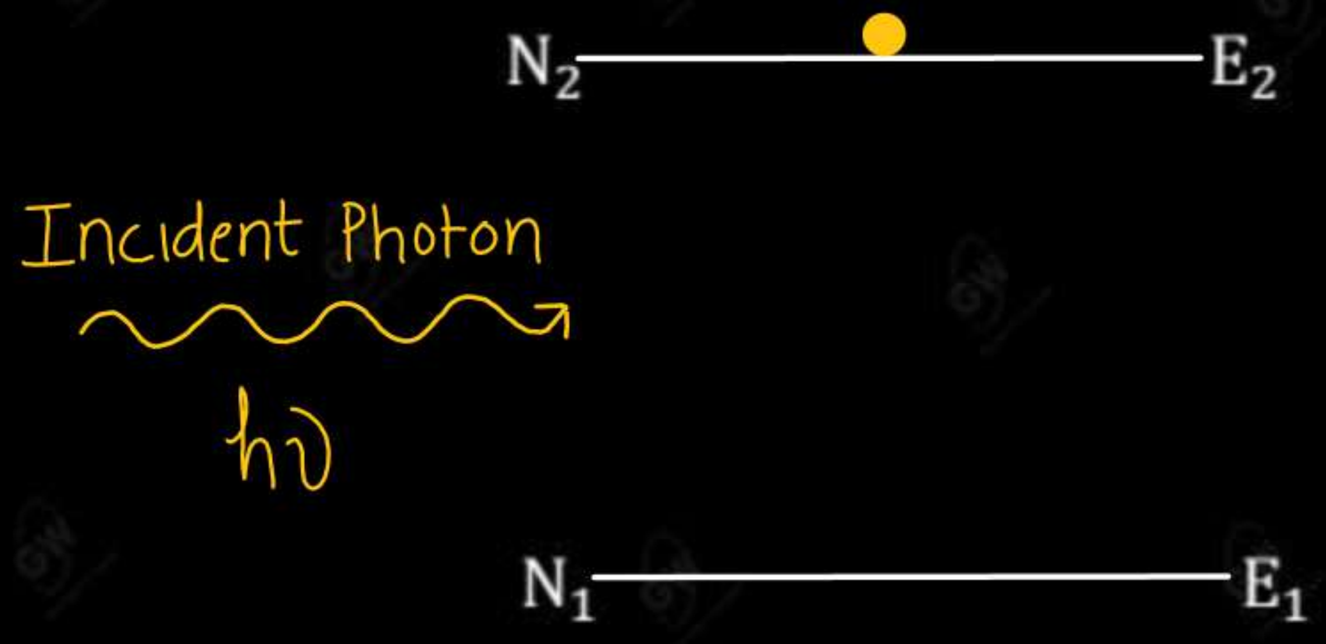
Where

A_{21} = Einstein's coefficient of spontaneous emission

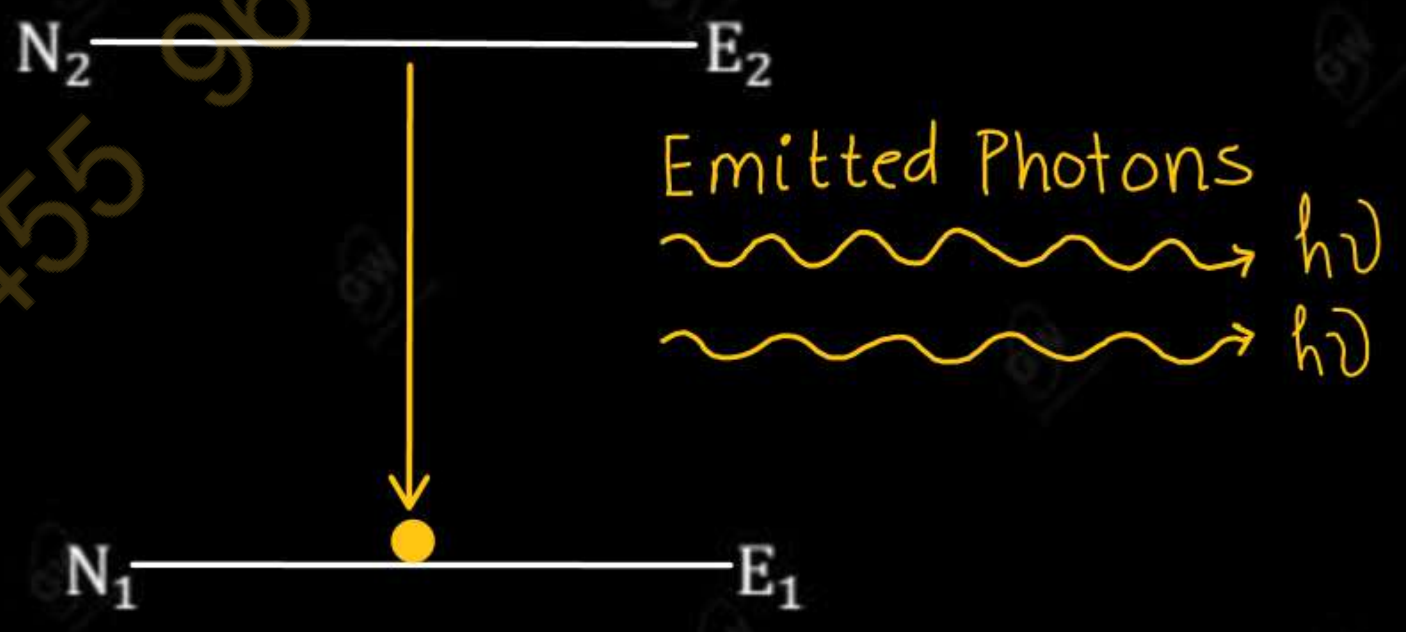
- Ordinary light is obtained from spontaneous emission

Stimulated Emission

- As suggested by Einstein's in 1917, an atom in excited state E_2 can also make transition to lower energy state E_1 when triggered by photon of energy $h\upsilon$.



Before Stimulated Emission



After Stimulated Emission

- This transition produces a second photon which is identical to incident photon with respect to frequency, phase and propagation direction. This process is called stimulated emission

- The rate of stimulated emission is directly proportional to the (i) number of atoms in excited state (N_2) and (ii) energy density $E(\upsilon)$

$$P''_{21} \propto N_2$$

$$P''_{21} \propto E(\upsilon)$$

$$P''_{21} \propto N_2 \quad E(\upsilon)$$

$$\boxed{P''_{21} = N_2 B_{21} E(\upsilon)}$$

Where

B_{21} = Einstein's coefficient of stimulated emission

- LASER light is obtained from stimulated emission

Difference between Spontaneous Emission and Stimulated Emission

Spontaneous Emission	Stimulated Emission
1. Emitted photons can move in any directions.	1. Emitted photons move in same directions.
2. Emitted radiations are not coherent.	2. Emitted radiations are coherent
3. Emitted photons from various atoms have no phase relationship between them	3. Emitted photons have same frequency and are in phase with incident photons. .
4. The rate of spontaneous emission is directly proportional to the number of atom in excited state (N_2)	4. The rate of stimulated emission is directly proportional to the number of atoms in excited state (N_2) and energy density $E(\upsilon)$
5. Ordinary light is obtained from spontaneous emission	5. LASER light is obtained from stimulated emission

Relation between Einstein's Coefficients

- Let N_1 and N_2 be the number of atoms at any instant in the lower energy State 1 and Higher energy State 2

- The rate of absorption is directly Proportional to the
  (i) The number of atom in state 1 (N_1)
  (ii) Energy Density $E(\nu)$

$$P_{12} \propto N_1$$

$$P_{12} \propto E(\nu)$$

$$P_{12} \propto N_1 E(\nu)$$

$$P_{12} = N_1 B_{12} E(\nu) \quad \text{——— (1)}$$

- The rate of spontaneous emission is directly proportional to the number of atoms in excited state (N_2)

$$P'_{21} \propto N_2$$

$$P'_{21} = N_2 A_{21} \quad \text{——— (2)}$$

- The rate of Stimulated emission is directly proportional to the

(i) Number of atoms in Excited State(N_2)

(ii) Energy Density $E(\nu)$

$$P''_{21} \propto N_2$$

$$P''_{21} \propto E(\nu)$$

$$P''_{21} \propto N_2 E(\nu)$$

$$P''_{21} = N_2 B_{21} E(\nu) \quad \text{——— (3)}$$

Under thermal Equilibrium

$$P_{12} = P'_{21} + P''_{21}$$

$$N_1 B_{12} E(\nu) = N_2 A_{21} + N_2 B_{21} E(\nu)$$

$$N_1 B_{12} E(\nu) - N_2 B_{21} E(\nu) = N_2 A_{21}$$

$$(N_1 B_{12} - N_2 B_{21}) E(\nu) = N_2 A_{21}$$

$$E(\nu) = \frac{N_2 A_{21}}{N_1 B_{12} - N_2 B_{21}}$$

$$E(\nu) = \frac{\cancel{N_2} A_{21}}{\cancel{N_2} B_{21}\left(\frac{B_{12}}{B_{21}} \times \frac{N_1}{N_2} - 1\right)}$$

$$E(\nu) = \frac{A_{21}}{B_{21}\left(\frac{B_{12}}{B_{21}} \times \frac{N_1}{N_2} - 1\right)} \quad \text{--- (4)}$$

According to Botlzmann's Distribution Law

$$\frac{N_1}{N_2} = e^{\frac{E_2 - E_1}{KT}}$$

$$\boxed{\frac{N_1}{N_2} = e^{\frac{h\nu}{KT}}} \quad \text{--- (5)}$$

V. imp

Using (5) in (4)

$$E(\nu) = \frac{A_{21}}{B_{21}\left(\frac{B_{12}}{B_{21}} \times e^{\frac{h\nu}{KT}} - 1\right)} \quad \text{--- (6)}$$

According to Planck's Radiation Law

$$E(\nu) = \frac{8\pi h\nu^3}{c^3} \times \frac{1}{\left(e^{\frac{h\nu}{KT}} - 1\right)} \quad \text{--- (7)}$$

Comparing (6) and (7)

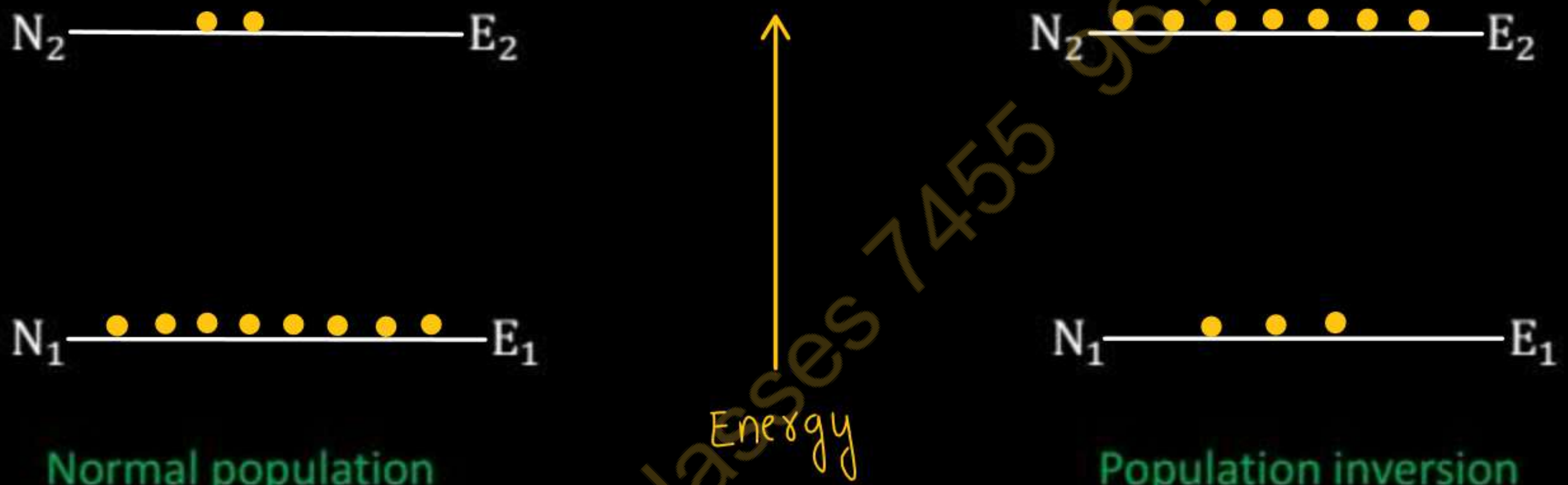
$$\boxed{\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3}}$$

$$\frac{B_{12}}{B_{21}} = 1$$

$$\boxed{B_{12} = B_{21}}$$

Population Inversion

- The situation in which the number of atoms in the higher energy state becomes comparatively greater than the number of atoms in the lower energy state is known as population inversion



Normal population

$N_1 > N_2$

Population inversion

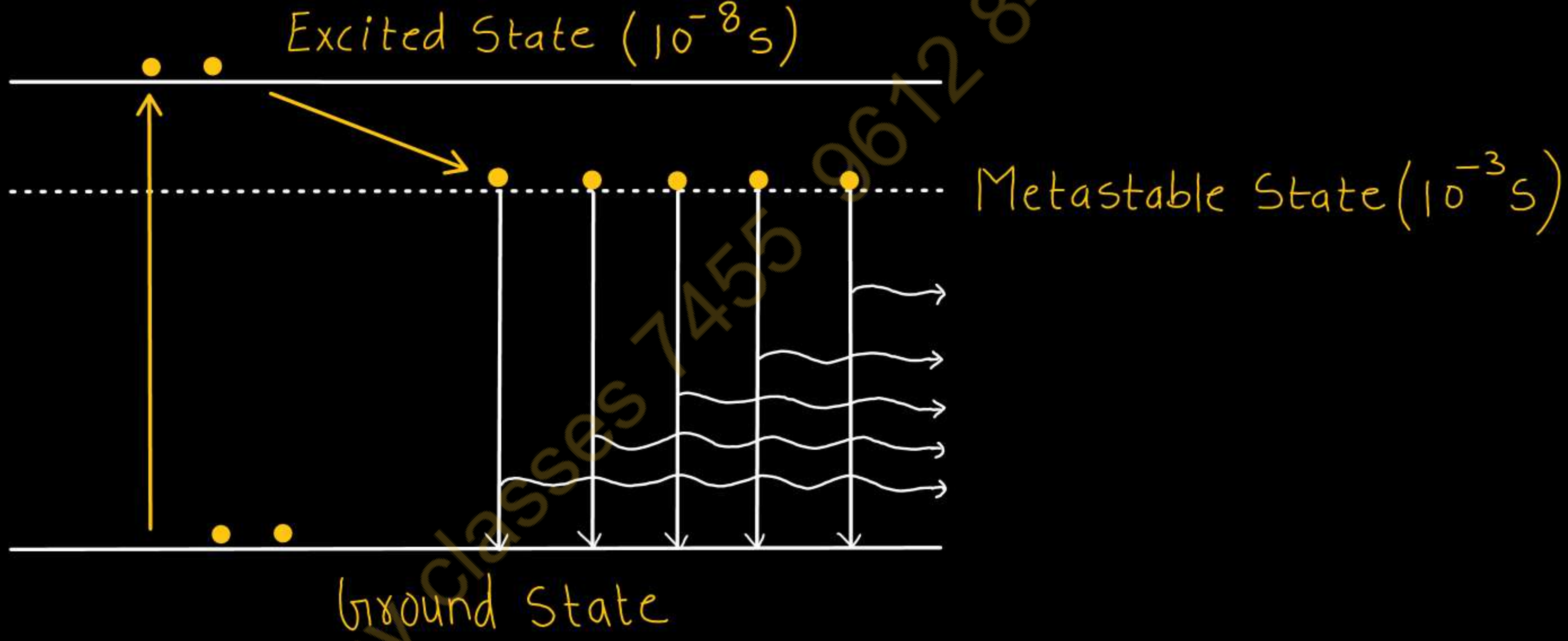
$N_1 < N_2$

- For laser action to take place, the higher energy level should be more populated than the lower energy level. i.e

$$N_2 > N_1$$

Metastable State

- Metastable State can be defined as a state where excited atom can remain for longer time than the normal excited state.



- It is also the excited state but having life time of 10^{-3} seconds.
- The population inversion occurs only in between the metable state and lower state.
- If metastable state do not exist, there could be no population inversion, no stimulated emission and hence no laser operation.

Necessary condition to achieve laser action

(i) The rate of emission must be greater than the rate of absorption

- Under normal condition, the number of atoms in the upper energy level is always smaller than that in lower energy level. If by some means, the number of atoms in higher energy state is made greater than that in lower energy state, the emission rate will become greater than the absorption rate.

(ii) The probability of spontaneous emission must be negligible in comparison to the probability of stimulated emission

- The condition can be achieved by taking working substance (active medium) such that its atoms have metastable states which have a lifetime 10^{-3} sec or more instead of the usual 10^{-8} sec. If certain atoms are excited to metastable state the probability of spontaneous emission will be quite negligible.

(ii) The coherent beam of light must be sufficiently amplified

- For this, active medium must be placed between two reflecting mirrors. The one of the mirror is fully reflecting while the other is partially transmitting.

Pumping

- The process by which we can raise the number of atoms from lower energy state to Higher state is known as pumping.

Two important method of pumping

(i) Optical Pumping

- In optical pumping the energy is supplied in the form of short flashes of light .
- The ground state atoms absorb this light energy and excite to higher energy state.
- This type of pumping is used in Ruby Laser

(ii) Electrical Pumping

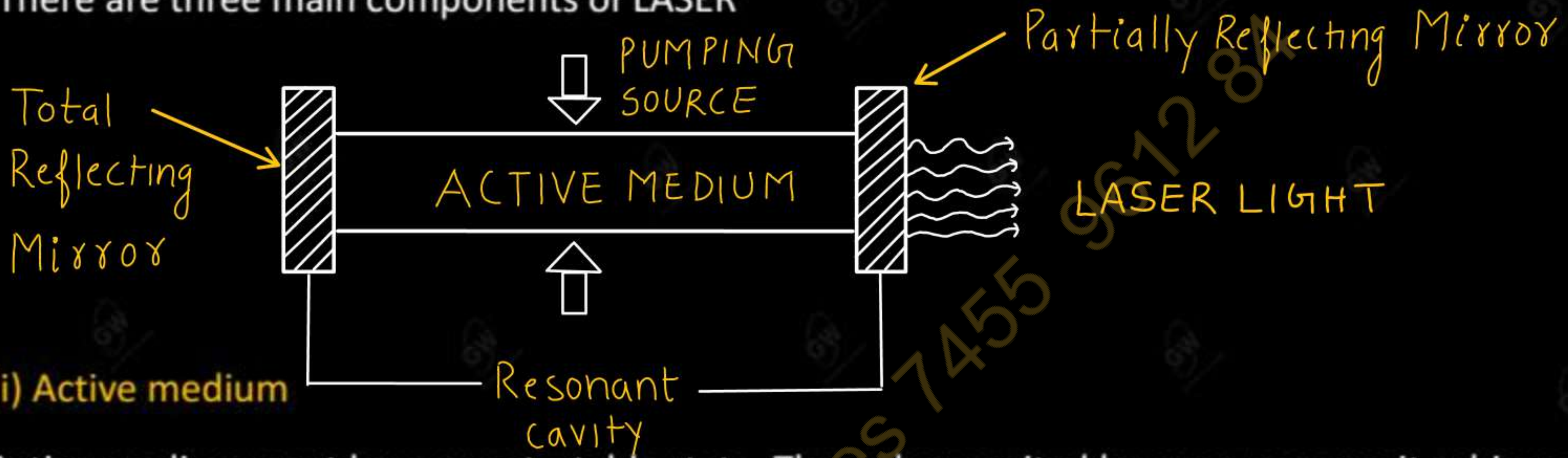
- In this method gas is ionized by means of suitable potential difference.
- Strong electric is applied to accelerate electrons emitted by cathode of the tube.
- During the motion towards the anode the accelerated electrons collide with the atoms of the medium and give up their energy to excite them to upper energy level.
- This type of pumping is used in Helium – Neon Laser

Principle of LASER

- All lasers work on the principle of "STIMULATED EMISSION" OF RADIATION with LIGHT AMPLIFICATION
- The ground state atom absorbs energy from external source and raise to higher state
- Due to short life (10^{-8} sec) of excited state, it spontaneously comes down to metastable state of long life (10^{-3} sec).
- Thus population inversion is achieved.
- A spontaneously emitted photon trigger other atoms of metastable state and initiating stimulated emission.
- The light beam is then amplified by the cavity resonator by multiple reflections between a pair of mirrors. This way, amplified laser beam is achieved.

Components of LASER

There are three main components of LASER



(i) Active medium

- Active medium must have a metastable state. Thus when excited by energy source it achieves population inversion.
- Active medium decide the types of laser:

  (i) Solid laser

  (ii) Liquid laser

  (iii) Gas laser

(ii) Pumping source (or Energy source)

- With the help of energy source the system can be raised to an excited state.
- With the help of this source the number of atoms in higher energy state may be increased and hence the population inversion is achieved.

(iii) Resonant Cavity or Optical Resonator

- The optical resonator consists of two reflecting mirrors R_1 and R_2
- The mirror R_1 is fully reflecting while the other mirror R_2 is partially reflecting
- Active medium is placed between R_1 and R_2
- These mirror act as an optical resonator or resonant cavity

EINSTEIN'S COEFFICIENTS

- The Einstein's coefficients are mathematical quantities describing the probability of absorption and emission of a photon by an atom or molecule

  There are three Einstein's Coefficients

(i) B_{12} : Einstein's coefficient for Stimulated Absorption

(ii) A_{21}: Einstein's coefficient for Spontaneous Emission

(iii) B_{21}: Einstein's coefficient for Stimulated Emission

Three Level Lasers

In this laser system

E_1 : Ground State (Known as lower lasing level)

E_2 : Metastable State (Known as upper lasing level)

E_3 : Excited State (Known as pumping level)

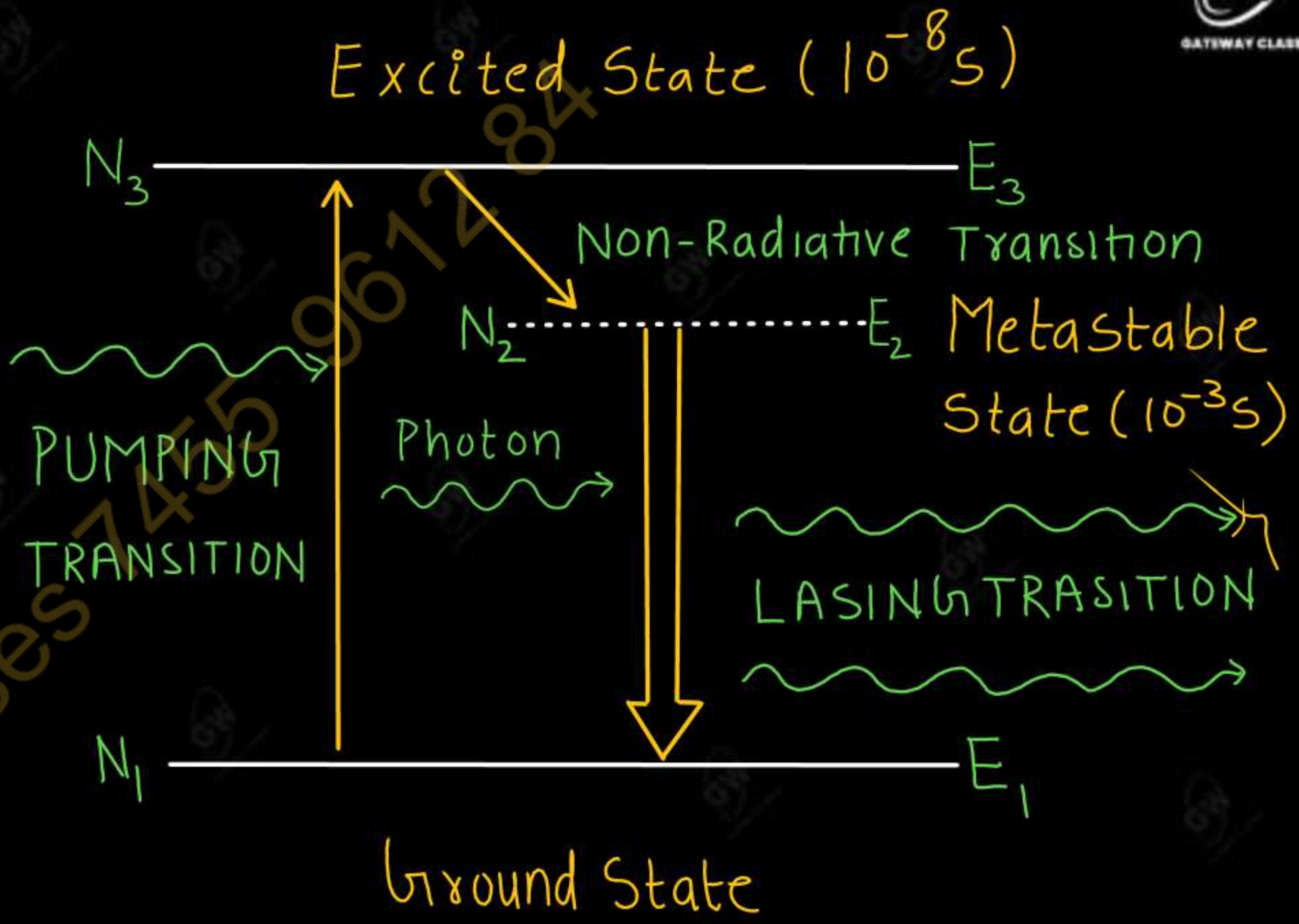
TRANSITIONS

1 - 3 : Pumping Transitions

3 - 2 : Non Radiative Transitions

2 - 1 : Spontaneous Emission

2 - 1 : Stimulated Emission



- ➢ The source of pumping excites some of the atoms from the ground level E_1 to a short lived (10^{-8}s) upper most pumping level E_3.

- Some of these atoms make spontaneous transition to the lowest energy level E_1 but most of them decay rapidly into an intermediate metastable state E_2. This transition from E_3 to E_2 is radiation less or non-radiative.
- As the energy level E_2 is long lived, the atomic population of metastable state E_2 goes on increasing gradually, while the atomic population of ground level E_1 goes on decreasing. Because of continuous pumping the population inversion is achieved between E_2 and E_1 levels.
- Now photon of energy $\mathrm{h}\nu = (E_2 - E_1)$ can initiate stimulated or laser transition.

Example : Ruby Laser

Disadvantage of three level laser : Difficult to produce population inversion because it requires very high pumping power

Four Level Lasers

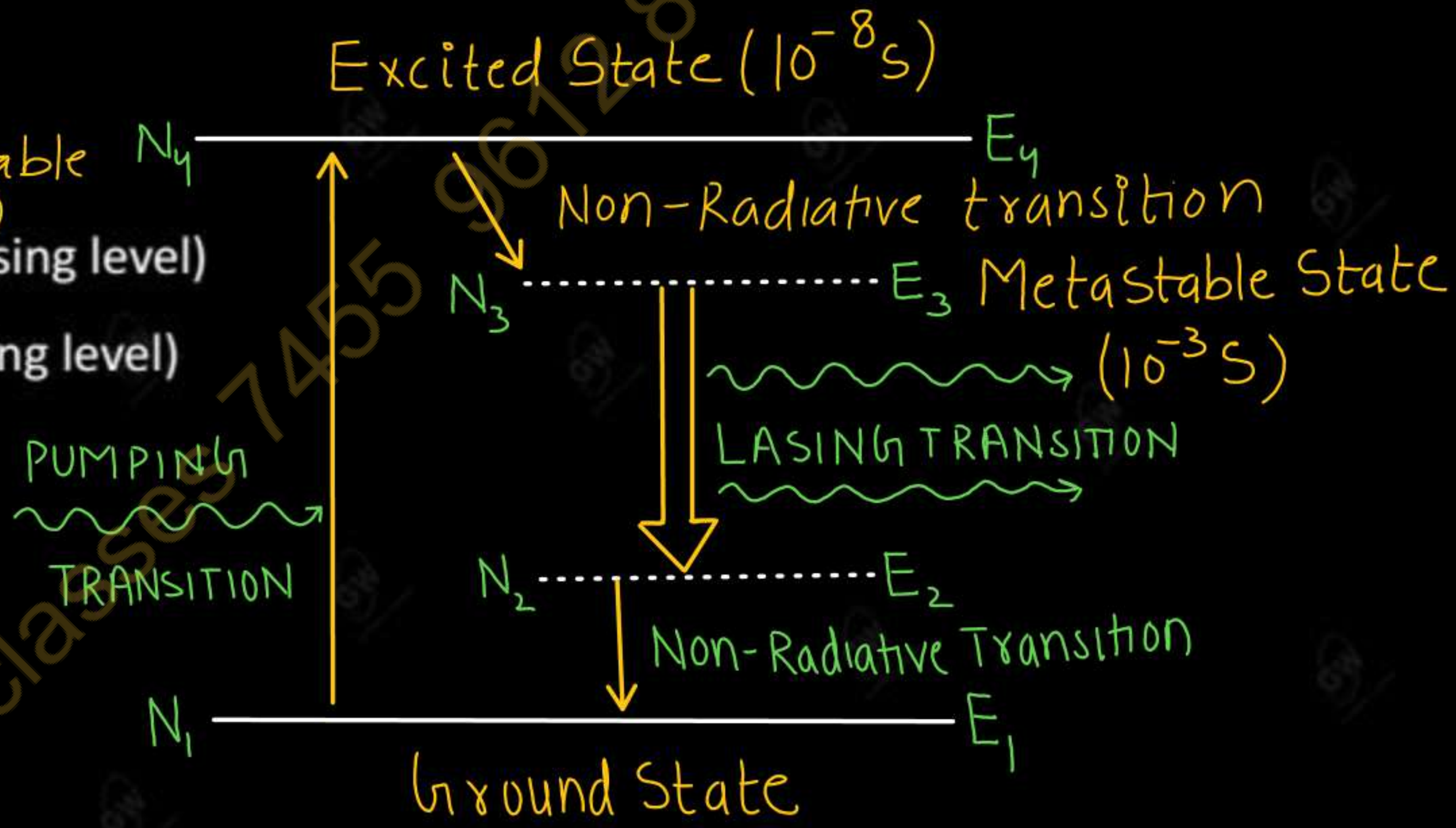
In this laser system

E_1 : Ground State

E_2 : Lower Lasing level (Lower Metastable State)

E_3 : Metastable State (or known as upper lasing level)

E_4 : Excited State (or known as pumping lasing level)



TRANSITIONS

1 - 4 : Pumping Transitions

4 - 3 : Non Radiative Transitions

3 - 2 : Spontaneous Emission

3 - 2 : Stimulated Emission

2 - 1 : Fast Transitions

- The source of pumping excites some of the atoms from the ground level E_1 to a short lived upper most pumping level E_4.
- The transitions $\mathrm{E_4}$ to $\mathrm{E_3}$ as well as $\mathrm{E_2}$ to $\mathrm{E_1}$ are radiation less transition and much faster as compared to $\mathrm{E_3}$ to $\mathrm{E_2}$
- As the lower lasing level E_2 is not the ground state, but very much above the ground level E_1, it is virtually vacant or free from any atoms. Due to continuous pumping, population inversion is achieved between the levels E_3 and E_2. A spontaneous photon of energy $\mathrm{h}\nu = E_3 - E_2$ can initiate a chain of stimulated emission.

Advantage of four level Laser over three level Laser

- The pumping power needed for the excitation of atoms is much lower than is a three level laser.
- The efficiency of four level laser is much better than that of a three level laser.
- Continuous operation is possible

Ruby Laser

- This is the first laser developed by T.H Maiman in 1960.
- The main characteristics of Ruby Laser are

(i) It is a solid state laser.

(ii) It is a three level laser.

(iii) Population inversion is achieved by using optical pumping.

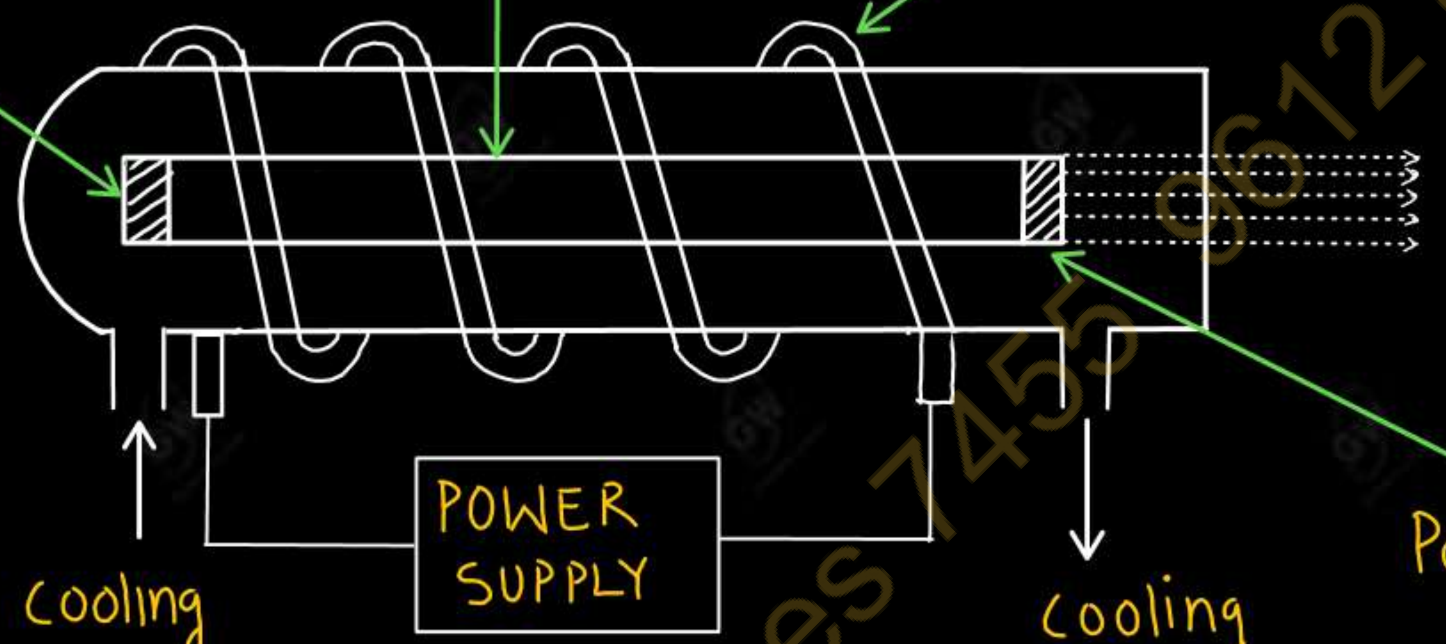
(iv) Output is in pulse form.

(v) External cooling is required

Construction

(i) Active Medium

- Ruby Rod is used as an active medium.
- Ruby rod is a crystal of aluminium oxide (Al_2O_3) doped with 0.05% of chromium oxide (Cr_2O_3).
- Some Al^{3+} ions are replaced by Cr^{3+} ions and these Cr^{3+} ions give pink colour to the ruby rod.
- Cr^{3+} ions act as active centres and responsible for laser action.



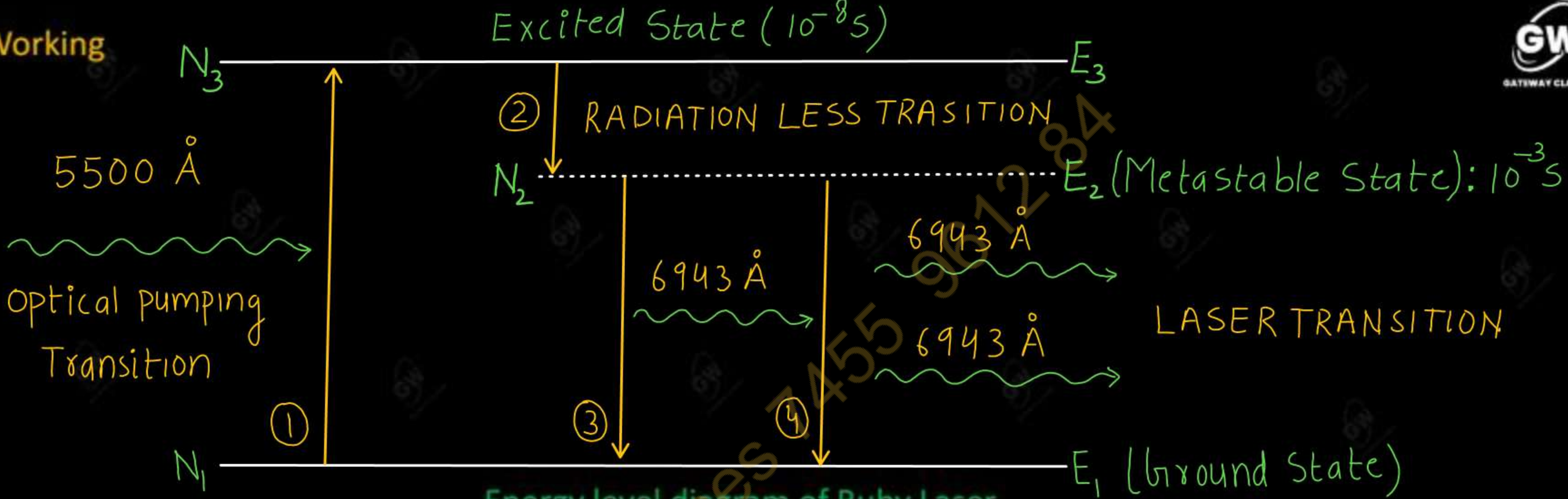
(ii) Pumping Source

- The Ruby rod is placed inside a helical shaped xenon flash lamp to excite the Cr^{3+} ions. Thus in ruby laser population inversion is achieved by using optical pumping.

(iii) Resonant Cavity

- The Ruby crystal is taken in the form of cylindrical rod and its ends are optically flat and parallel.
- One end of the ruby rod is coated with silver completely while the other one is partially silvered.
- These two silver coated ends of the ruby rod act as a Resonant Cavity.

Working



Energy level diagram of Ruby Laser

- Ruby Laser is a three level laser system and its energy level diagram consist of

  E_1 : Ground state

  E_2 : Metastable state

  E_3 : Excited state

- The energy difference $E_3 - E_1$ corresponds to a wave length of 5500 Å.

- Most of the Chromium ions are in the ground state E_1. After absorbing light (Photons) of wavelength 5500 Å from xenon flash lamp, some of the Cr^{3+} ions at ground energy level E_1 jump to higher energy level E_3. This is known as Pumping transition
- At Higher energy level (E_3), Cr^{3+} ions are unstable and by loosing a part of their energy to the crystal lattice, they fall metastable state E_2. This is known as Non-radiative transition.
- Since E_2 has a much longer life time (about 10^{-3} s), the number of Cr^{3+} ions goes on increasing in E_2 state while the number of these ions in ground state E_1 goes on decreasing due to pumping by the flash lamp. Thus population inversion is established between the metastable state E_2 and ground state E_1.
- Now, some of the Cr^{3+} ions will fall spontaneously to the ground state E_1 by emitting photons of wavelength 6943 Å. This is known as Spontaneous Emission
- The photons that are moving parallel to the axis of the rod will reflect back and forth by the silvered ends of the rod and stimulate other excited Cr^{3+} ions to radiate another photons with the same phase. This is known as Stimulated Emission

➢ Thus, due to successive reflection of these photons at the ends of the rod, the number of photons multiply. After a few microseconds, a monochromatic, intense, coherent, unidirectional and collimated beam of red light of wavelength 6943 Å emerges through the partially silvered end of the rod. Output is in pulse form.

Drawbacks

(i) The output of LASER is not continuous and occurs in the form of pulse.

(ii) It requires greater excitation in order to achieve population inversion.

(iii) Low efficiency.

(IV) External cooling is required

Helium-Neon Laser

- Ali Javan invented He-Ne gas laser in 1960. It is the first gas laser which was operated successfully.
- Main characteristics of He-Ne Laser

(i) It is a gas laser

(ii) It is a four level laser

(iii) Population inversion is achieved by using electrical pumping.

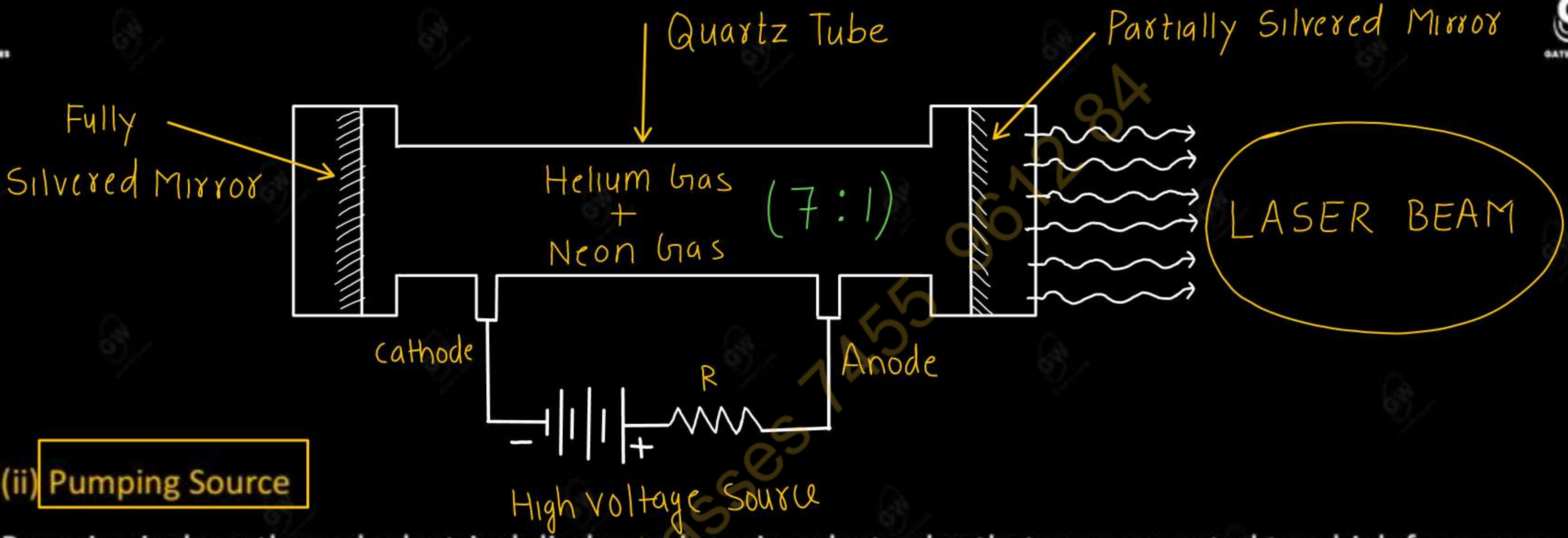
(iv) Output is in continuous form.

(v) External cooling is not required

Construction

(i) Active Medium

- He-Ne gas laser consist of a narrow quartz tube filled with a mixture of Helium and Neon in the ratio of 7:1 respectively at low pressure.
- Ne atoms act as active centres and responsible for the laser action while He atoms are used to help in the excitation process.



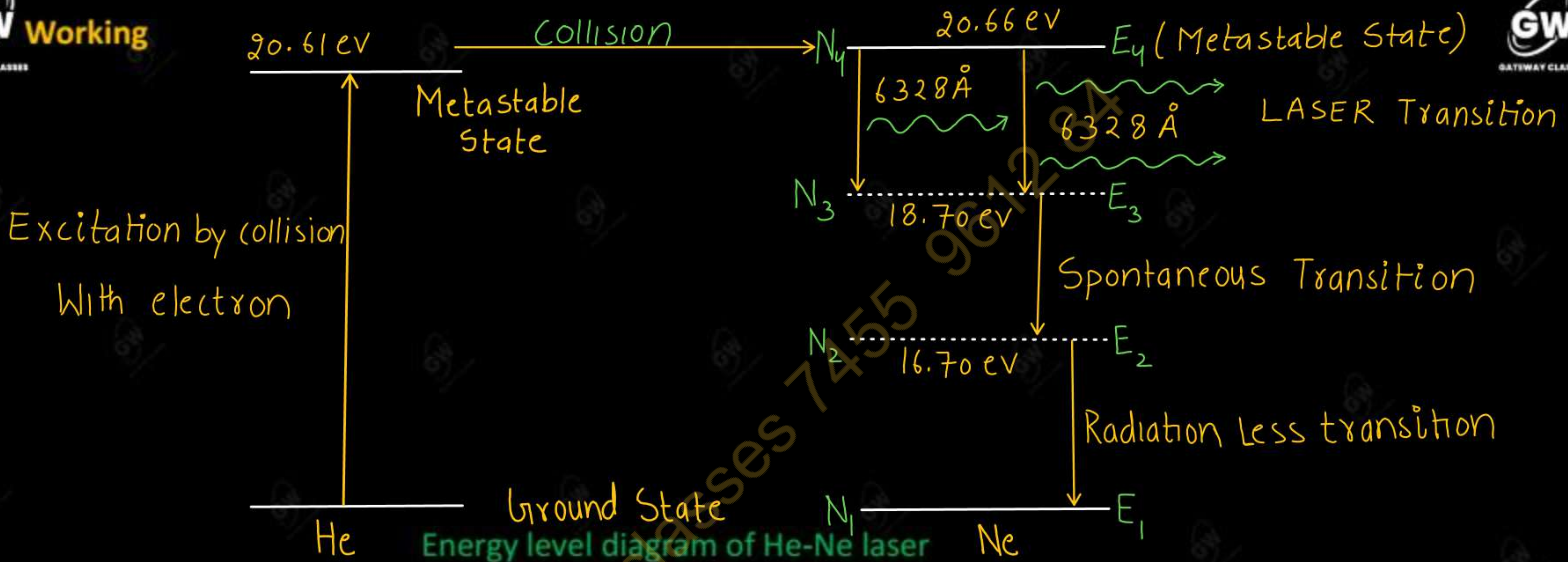
(ii) Pumping Source

- Pumping is done through electrical discharge by using electrodes that are connected to a high frequency alternating source.

(iii) Resonant Cavity

- To construct Resonant Cavity, two parallel mirrors are placed at the ends of the quartz tube one of them is partially reflecting while the other is fully reflecting.

Working



Energy level diagram of He-Ne laser

- He – Ne Laser is a four energy level laser system. The electrons produced from electric discharge collide with He and Ne atom and Excite or pump them to higher energy level (Metastable state) at 20.61 ev and 20.66 ev respectively.

As He atoms are lighter than the Ne atoms, they are easily excite to metastable state at 20.61 ev.

Some of the excited He atoms transfer their energy to the ground state Ne atoms by collision, with 0.5 ev of additional energy being provided by the K.E. of atoms.

$$He + e \rightarrow He^{*}$$

$$He^{*} + Ne \rightarrow Ne^{*} + He$$

Thus He atoms help in achieving a population inversion in Ne atoms.

- When an exited Ne-atoms drop down spontaneously from metastable state at 20.66 ev to lower energy state at 18.70 ev, it emits a 6328 Å photon in visible region.
- This photon travels through the mixture of gas and if it is moving parallel to the axis of tube is reflected back and forth motion by reflector ends until it stimulates an excited Ne-atom and causes it to emit a fresh 6328 Å photon in phase with stimulating photon. The stimulated transition from 20.66 ev level to 18.70 ev level is the laser transition.

- Thus, due to successive reflections of these photons at the ends of the tube the number of photon multiplies. After a few microseconds a monochromatic, intense, coherent, unidirectional and collimated beam of red light of wavelength 6328 Å emerges through the partially silvered mirror.
- The Ne-atoms drop down from 18.70 ev to lower metastable state through spontaneous emission by emitting in coherent light.
- From level E_2 Ne atoms come to ground state by losing their remaining energy through collision with the tube walls. Hence, final transition is radiation less.

Applications of Laser

- In medicine laser beam is used in treatment of kidney stone, cancer, eye – surgery, cornea etc.
- The laser beam is used for drilling, welding and melting of hard materials like diamonds, iron, steel, etc.
- It is used in heat treatments for hardening or annealing in metallurgy.
- Laser is used in holography and fiber optics
- Laser is very useful in science and research areas.
- Laser is used for communications and measuring large distances.
- Semiconductor lasers is used for recording and erasing of data on compact disks.
- Semiconductor laser and helium-neon lasers are used to scan the universal barcodes to identify products in supermarket scanners.
- During war-time, lasers are used to detect and destroy enemy missiles. Now, laser-pistols, laser-rifles and laser bombs are also being made, which can be aimed at the enemy in the night.

Q. 1 In a Ruby laser, total number of Cr^{3+} ions is 2.8×10^{19}. If the laser emits radiation of wavelength 7000Å, then calculate the energy of the laser pulse.

$$\boxed{\nu = \frac{c}{\lambda}}$$

Given

Number of Cr^{3+} ions(n)

$n = 2.8 \times 10^{19}$

$\lambda = 7000\,\text{Å} = 7000 \times 10^{-10}\,m$

$h = 6.62 \times 10^{-34}\,J\text{-}s$

$c = 3 \times 10^{8}\,m/s$

We know that

Energy of Laser Pulse (E) = Number of ions × Energy of one photon

$$E = nh\nu$$

$$E = \frac{nhc}{\lambda}$$

$$E = \frac{2.8 \times \cancel{10^{19}} \times 6.62 \times \cancel{10^{-34}} \times 3 \times \cancel{10^{8}}}{7 \times \cancel{10^{-7}}}$$

$$\boxed{E = 7.94\,J}$$

Q.2 Calculate the population ratio of two states in He – Ne laser that producers light of wavelength 6000Å at 300K.

Given

$\lambda = 6000\,\text{Å} = 6000 \times 10^{-10}\,m$

$T = 300\,K$

K = Boltzmann's constant

$K = 8.6 \times 10^{-5}\,eV/K$

$K = 8.6 \times 10^{-5} \times 1.6 \times 10^{-19}\,J/K$

We know that

$$\frac{N_2}{N_1} = e^{\frac{-(E_2-E_1)}{KT}} = e^{\frac{-h\nu}{KT}}$$

$$\frac{N_2}{N_1} = e^{\frac{-hc}{\lambda KT}}$$

$$\frac{N_2}{N_1} = e^{\frac{-6.62 \times \cancel{10^{-34}} \times 3 \times \cancel{10^{8}}^{\,3}}{6\cancel{000} \times \cancel{10^{-10}} \times 8.6 \times \cancel{10^{-5}} \times 1.6 \times \cancel{10^{-19}} \times 3\cancel{00}}}$$

$$\boxed{\frac{N_2}{N_1} = e^{-80}}$$

Q.3 Calculate the energy and momentum of a photon of a laser beam of wavelength 6328 Å

Given

$\lambda = 6328\,Å$

$h = 6.62 \times 10^{-34}\ J\text{-}S$

$c = 3 \times 10^{8}\ m/s$

Energy of photon (E)

$E = h\nu$

$$E = \frac{hc}{\lambda}$$

$$E = \frac{6.62 \times 10^{-34} \times 3 \times 10^{8}}{6328 \times 10^{-10}}$$

$$\boxed{E = 3.14 \times 10^{-19}\ J}$$

$$E = \frac{3.14 \times 10^{-19}}{1.6 \times 10^{-19}}$$

$$\boxed{E = 1.962\ eV}$$

Momentum of Photon (P)

$$P = \frac{h}{\lambda}$$

$$P = \frac{6.62 \times 10^{-34}}{6328 \times 10^{-10}}$$

$$\boxed{P = 1.05 \times 10^{-27}\ kgm/s}$$

LASER (DPP-2)

Q.1 What is the principle of laser?

Q.2 Difine Population inversion in LASER

Q.3 Write the essential requirements for laser action.

Q.3 Define metastable state.

Q.2 Differentiate between spontaneous and stimulated emission of radiation. Which one is required for laser action?

Q.4 What are Einstein's coefficients? Obtain a relation between them.

Q.5 What do you understand by three and four level lasers? What is the advantage of three level laser over four level laser?

Q.6 What are solid state lasers? Explain the construction and working of Ruby laser with suitable diagram. Compare if with He – Ne laser?

Q.7 Draw a neat diagram of He – Ne laser and describe its method of working. How it is superior to Ruby Laser ?

Q.8 Discuss important applications of laser.

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Thank You